

Comparisons of different AI techniques on the game 2048

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ULB
INFO-H410

Abstract

GitHub Repository —

<https://github.com/Somsro/infoH410>

Introduction

In the course INFO-H410 Techniques of Artificial Intelligence at ULB, students are required to apply and compare different AI techniques to the same problem. The game chosen is 2048. 2048 is a single-player game on a 4×4 grid where the player has to combine tiles by shifting them in one of the four possible directions: up, down, left and right. Each turn, a new tile with a value of 2 or 4 appears in an empty spot on the board. The player wins by creating a tile with the value of 2048, but can continue playing to reach higher values. The game ends when there are no more valid moves available.

The randomness of the games makes it a good candidate for testing the performance of different AI techniques. Three agents will be compared; Deep qlearner, Expectimax and an n-tuple network TD learning agent. The goal will be to evaluate their performance (the final score reached) but also the computational cost to better understand the trade-offs between planning and learned policies.

Problem Formulation

The game 2048 is a sequential decision making problem with uncertainty. The state of the game is represented by the configuration of the board. The configuration of the board is a 4×4 grid where each cell can be empty or contain a tile with a value that is a power of 2. At each time step, the agent observes the current board configuration and selects in which direction to shift the tiles in one of the four possible directions: up, down, left and right as long as there is at least one tile moving. Two tiles with the same value that collide during a shift will merge into a tile with the value of their sum. A new tile with a value of 2 or 4 will then appear in a random empty spot on the board which makes future states only partially controllable by the agent. The objective

of the agents is to achieve the highest score. The game ends when there are no more valid moves available. The fact that a new tile appears in a random empty spot on the board after each move makes the game stochastic and thus making it a good candidate for testing the performance of different AI techniques.

Formal Representation:

State Space (S): Values and grid positions of all tiles.

Actions (A): $\{Up, Down, Left, Right\}$, restricted to moves that modify the board.

Reward (r): Can be constructed based on several features (empty cells, monotonicity, and max tile placement).

Constraints: Terminal state is reached when no legal actions remain. Tabular Q-learning and exhaustive search are infeasible due to the massive state space.

Probabilities: Stochastic transitions $P(s' | s, a)$ dictate the game dynamics, preventing exact deterministic planning.

Methodologies

Three agents will be compared; Deep Q-Learning, Expectimax and an n-tuple network TD(0) learning agent. The three methods have each their own field of application and are based on different principles. Expectimax is a planning-based approach, deep Q-Learning is a deep reinforcement learning method that learns action values from experience meanwhile TD learning with function approximation is a lighter value-based alternative. Comparing them on the same problem will allow us to better understand the trade-offs between planning, learning, the computational cost as well as performance of each method. For each of the following approach, the states is encoded as a list of 16 integers, each representing the exponent k such that the tile value in that cell is 2^k . (For example, 16 is encoded as 4 since $2^4 = 16$ with the particularity that 0 represents an empty cell). The action space is the same for all agents and consists of the 4 possible moves: up, down, left and right encoded as integer values from 0 to 3. **Approach: Expectimax**

Approach: Deep Q-Learning (DQN)

Logic: Approximate the action value function (Q-value) using a deep neural network to bypass the exhaustive exploration of the state space. The agent learns directly from experience via stochastic gradient descent on (s, a, r, s')

tuples, guided by a shaped reward system reflecting 2048 heuristics (tile merges, empty cells, monotonicity, and corner penalties).

Assumption: The environment features a massive state space and stochastic transitions, rendering tabular methods infeasible. While this model-free approach is good at capturing complex patterns, it assumes access to high computational power and massive amounts of training data (very long training sessions) to converge.

Approach: N-tuple network TD(0) learning agent

Temporal-Difference learning with value function approximation provides a lighter alternative to Deep Q-Learning. Instead of learning action values directly, this approach estimates the value of board after-states and updates this estimate by bootstrapping from successor after-states. In this implementation, the after-states represent the move without placing a new tile yet in order to estimate the value function. An error signal is computed based on the difference between the current value estimate and the observed reward plus the value of the next after-state. The value function is approximated using an N-tuple network composed of lookup tables defined over several tile patterns and their symmetric variants.

Algorithm 1: Training procedure of the TD agent

Require: Number of episodes N

```

1: Initialize TD agent with N-tuple network  $V\theta$ 
2: for episode = 1 to  $N$  do
3:   Reset the environment
4:   Initialize empty trajectory path
5:   done  $\leftarrow$  false
6:   while not done do
7:     Select an action using  $\epsilon$ -greedy exploration
8:     Clone the environment and apply the action to
       obtain the after-state and reward
9:     Execute the action in the real environment
10:    Store the after-state, reward, and terminal flag in
       the path
11:    end while
12:     $target \leftarrow 0$ 
13:    for each stored after-state  $s$  in reverse order do
14:       $v \leftarrow V\theta(s)$ 
15:       $\delta \leftarrow target - v$ 
16:      Update the weights of  $V\theta$  using  $\delta$ 
17:      Update  $target \leftarrow r + V\theta(s)$ 
18:    end for
19:    Update  $\epsilon \leftarrow \max(\epsilon_{\min}, \epsilon \cdot \lambda)$ 
20: end for return trained agent

```

Using L^AT_EX to Format Your Paper

The latest version of the AAAI style file is available on AAAI’s website. Download this file and place it in the T_EX search path. Placing it in the same directory as the paper should also work. You must download the latest version of the complete AAAI Author Kit so that you will have the latest instruction set and style file.

Document Preamble

In the L^AT_EX source for your paper, you **must** place the following lines as shown in the example in this subsection. This command set-up is for three authors. Add or subtract author and address lines as necessary, and uncomment the portions that apply to you. In most instances, this is all you need to do to format your paper in the Times font. The helvet package will cause Helvetica to be used for sans serif. These files are part of the PSNFSS2e package, which is freely available from many Internet sites (and is often part of a standard installation).

Leave the setcounter for section number depth commented out and set at 0 unless you want to add section numbers to your paper. If you do add section numbers, you must uncomment this line and change the number to 1 (for section numbers), or 2 (for section and subsection numbers). The style file will not work properly with numbering of subsections, so do not use a number higher than 2.

The Following Must Appear in Your Preamble

```

\documentclass[letterpaper]{article}
% DO NOT CHANGE THIS
\usepackage{aaai2026} % DO NOT CHANGE THIS
\usepackage{times} % DO NOT CHANGE THIS
\usepackage{helvet} % DO NOT CHANGE THIS
\usepackage{courier} % DO NOT CHANGE THIS
\usepackage[hyphens]{url} % DO NOT CHANGE THIS
\usepackage{graphicx} % DO NOT CHANGE THIS
\urlstyle{rm} % DO NOT CHANGE THIS
\def\UrlFont{\rm} % DO NOT CHANGE THIS
\usepackage{graphicx} % DO NOT CHANGE THIS
\usepackage{natbib} % DO NOT CHANGE THIS
\usepackage{caption} % DO NOT CHANGE THIS
\frenchspacing % DO NOT CHANGE THIS
\setlength{\pdfpagewidth}{8.5in} % DO NOT CHANGE THIS
\setlength{\pdfpageheight}{11in} % DO NOT CHANGE THIS
%
% Keep the \pdfinfo as shown here. There’s no need
% for you to add the /Title and /Author tags.
\pdfinfo{
  /TemplateVersion (2026.1)
}

```

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After the preamble above, you should prepare your paper as follows:

```

\begin{document}
\maketitle
\begin{abstract}
%...
\end{abstract}

```

If you want to add links to the paper’s code, dataset(s), and extended version or similar this is the place to add them, within a *links* environment:

```

\begin{links}
\link{Code}{https://aaai.org/example/guidelines}
\link{Datasets}{https://aaai.org/example/datasets}
\link{Extended version}{https://aaai.org/example}
\end{links}

```

You should then continue with the body of your paper. Your paper must conclude with the references, which should be inserted as follows:

```
% References and End of Paper
% These lines must be placed at the end of your paper
\bibliography{Bibliography-File}
\end{document}

\begin{document}
\maketitle
...
\bibliography{Bibliography-File}
\end{document}
```

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Papers must be formatted to print in two-column format on 8.5 x 11 inch US letter-sized paper. The margins must be exactly as follows:

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Column Width and Margins. To ensure maximum readability, your paper must include two columns. Each column should be 3.3 inches wide (slightly more than 3.25 inches), with a .375 inch (.952 cm) gutter of white space between the two columns. The `aaai2026.sty` file will automatically create these columns for you.

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Commands that alter page layout are forbidden. These include `\columnsep`, `\float`, `\topmargin`, `\topskip`, `\textheight`, `\textwidth`, `\oddsidemargin`, and `\evensidemargin` (this list is not exhaustive). If you alter page layout, you will be required to pay the page fee. Other commands that are questionable and may cause your paper to be rejected include `\parindent`, and `\parskip`. Commands that alter the space between sections are forbidden. The title sec package is not allowed. Regardless of the above, if your paper is obviously "squeezed" it is not going to be accepted. Options for reducing the length of a paper include reducing the size of your graphics, cutting text, or paying the extra page charge (if it is offered).

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Do not use type 3 fonts for any portion of your paper, including graphics. Type 3 bitmapped fonts are designed for fixed resolution printers. Most print at 300 dpi even if the printer resolution is 1200 dpi or higher. They also often cause high resolution imagesetter devices to crash. Consequently, AAAI will not accept electronic files containing obsolete type 3 fonts. Files containing those fonts (even in graphics) will be rejected. (Authors using blackboard symbols must avoid packages that use type 3 fonts.)

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<code>\abovecaption</code>	<code>\abovedisplay</code>	<code>\addevensidemargin</code>	<code>\addsidemargin</code>
<code>\addtolength</code>	<code>\baselinestretch</code>	<code>\belowcaption</code>	<code>\belowdisplay</code>
<code>\break</code>	<code>\clearpage</code>	<code>\clip</code>	<code>\columnsep</code>
<code>\float</code>	<code>\input</code>	<code>\input</code>	<code>\linespread</code>
<code>\newpage</code>	<code>\pagebreak</code>	<code>\renewcommand</code>	<code>\setlength</code>
<code>\text height</code>	<code>\tiny</code>	<code>\top margin</code>	<code>\trim</code>
<code>\vskip{-</code>	<code>\vspace{-</code>		

Table 1: Commands that must not be used

<code>authblk</code>	<code>babel</code>	<code>cjk</code>	<code>dvips</code>
<code>epsf</code>	<code>epsfig</code>	<code>euler</code>	<code>float</code>
<code>fullpage</code>	<code>geometry</code>	<code>graphics</code>	<code>hyperref</code>
<code>layout</code>	<code>linespread</code>	<code>lmodern</code>	<code>maltepaper</code>
<code>navigator</code>	<code>pdfcomment</code>	<code>pgfplots</code>	<code>psfig</code>
<code>pstricks</code>	<code>tlenc</code>	<code>titlesec</code>	<code>tocbind</code>
<code>ulem</code>			

Table 2: LaTeX style packages that must not be used.

The default size for your type must be ten-point with twelve-point leading (line spacing). Start all pages (except the first) directly under the top margin. (See the next section for instructions on formatting the title page.) Indent ten points when beginning a new paragraph, unless the paragraph begins directly below a heading or subheading.

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Formatting Author Information. Author information has to be set according to the following specification depending if you have one or more than one affiliation. You may not use a table nor may you employ the `\authorblk.sty` package. For one or several authors from the same institution, please separate them with commas and write all affiliation directly below (one affiliation per line) using the macros `\author` and `\affiliations`:

```
\author{
  Author 1, ..., Author n\\
}
\affiliations {
  Address line\\
  ... \\
  Address line\\
}
```

For authors from different institutions, use `\textsuperscript{\rm x}` to match authors and affiliations. Notice that there should not be any spaces between the author name and the superscript (and the comma should come after the superscripts).

```
\author{
  AuthorOne\equalcontrib\textsuperscript{\rm 1,\rm2},
  AuthorTwo\equalcontrib\textsuperscript{\rm 2},
  AuthorThree\textsuperscript{\rm 3},\\
  AuthorFour\textsuperscript{\rm 4},
  AuthorFive\textsuperscript{\rm 5}}
\affiliations {
  \textsuperscript{\rm 1}AffiliationOne,\\
  \textsuperscript{\rm 2}AffiliationTwo,\\
  \textsuperscript{\rm 3}AffiliationThree,\\
  \textsuperscript{\rm 4}AffiliationFour,\\
  \textsuperscript{\rm 5}AffiliationFive\\
  \{email, email\}@affiliation.com,
  email@affiliation.com,
  email@affiliation.com,
  email@affiliation.com
}
```

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\title{Very Important Results in AI}\thanks{This work is supported by everybody.}}
```

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Citations within the text should include the author's last name and year, for example (Newell 1980). Append lower-case letters to the year in cases of ambiguity. Multiple authors should be treated as follows: (Feigenbaum and Englemore 1988) or (Ford, Hayes, and Glymour 1992). In the case of four or more authors, list only the first author, followed by et al. (Ford et al. 1997).

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Long quotations and extracts should be indented ten points from the left and right margins.

This is an example of an extract or quotation. Note the indent on both sides. Quotation marks are not necessary if you offset the text in a block like this, and properly identify and cite the quotation in the text.

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Use footnotes judiciously, taking into account that they interrupt the reading of the text. When required, they should be consecutively numbered throughout with superscript Arabic

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When necessary, headings should be used to separate major sections of your paper. Remember, you are writing a short paper, not a lengthy book! An overabundance of headings will tend to make your paper look more like an outline than a paper. The `aaai2026.sty` package will create headings for you. Do not alter their size nor their spacing above or below.

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Section Headings. Sections should be arranged and headed as follows:

1. Main content sections
2. Appendices (optional)
3. Ethical Statement (optional, unnumbered)
4. Acknowledgements (optional, unnumbered)
5. References (unnumbered)

Appendices. Any appendices must appear after the main content. If your main sections are numbered, appendix sections must use letters instead of arabic numerals. In $\LaTeX$ you can use the `\appendix` command to achieve this effect and then use `\section{Heading}` normally for your appendix sections.

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Acknowledgments. The acknowledgments section, if included, appears right before the references and is headed "Acknowledgments". It must not be numbered even if other sections are (use `\section*{Acknowledgements}` in $\LaTeX$). This section includes acknowledgments of help from associates and colleagues, credits to sponsoring agencies, financial support, and permission to publish. Please acknowledge other contributors, grant support, and so forth, in this section. Do not put acknowledgments in a footnote on the first page. If your grant agency requires acknowledgment of the grant on page 1, limit the footnote to the required statement, and put the remaining acknowledgments at the back. Please try to limit acknowledgments to no more than three sentences.

References. The references section should be labeled "References" and must appear at the very end of the paper (don't end the paper with references, and then put a figure by itself on the last page). A sample list of references is given later on in these instructions. Please use a consistent format for references. Poorly prepared or sloppy references reflect

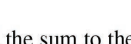
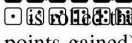
if there are no  show  add the sum to their turn total. At each decision point, a player may continue to roll or stop. If they decide to stop, they add their turn total to their total score and then it becomes the opponent's turn. Otherwise, they roll dice again  continue adding to their turn total. If a single  turn  and the turn ended (no points gained); if a  then the players

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Figures, drawings, tables, and photographs should be placed throughout the paper on the page (or the subsequent page) where they are first discussed. Do not group them together at the end of the paper. If placed at the top of the paper, illustrations may run across both columns. Figures must not invade the top, bottom, or side margin areas. Figures must be inserted using the `\usepackage{graphicx}`. Number figures sequentially, for example, figure 1, and so on. Do not use minipage to group figures.

If you normally create your figures using pgfplots, please create the figures first, and then import them as pdfs with proper bounding boxes, as the bounding and trim boxes created by pgfplots are fragile and not valid.

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`\shortcite`: Cites the given reference(s) with just the year. This appears as “(Year)” for one reference, or “(Year; Year)” for multiple references.

`\citeauthor`: Cites the given reference(s) with just the author name(s) and no parentheses.

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Please check all the pages of your PDF file. The most commonly forgotten element is the acknowledgements — especially the correct grant number. Authors also commonly forget to add the metadata to the source, use the wrong reference style file, or don’t follow the capitalization rules or comma placement for their author-title information properly. A final common problem is text (especially equations) that runs into the margin. You will need to fix these common errors before submitting your file.

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If you have any questions about the preparation or submission of your paper as instructed in this document, please contact AAAI Press at the address given below. If you have technical questions about implementation of the `aaai` style file, please contact an expert at your site. We do not provide technical support for L^AT_EX or any other software package. To avoid problems, please keep your paper simple, and do not incorporate complicated macros and style files.

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Additional Resources

L^AT_EX is a difficult program to master. If you’ve used that software, and this document didn’t help or some items were not explained clearly, we recommend you read Michael Shell’s excellent document (`testflow doc.txt V1.0a 2002/08/13`) about obtaining correct PS/PDF output on L^AT_EX systems. (It was written for another purpose, but it has general application as well). It is available at www.ctan.org in the `tex-archive`.

Reference Examples

Formatted bibliographies should look like the following examples. You should use BibTeX to generate the references. Missing fields are unacceptable when compiling references, and usually indicate that you are using the wrong type of entry (BibTeX class).

Book with multiple authors Use the `@book` class.

Engelmore, R.; and Morgan, A., eds. 1986. *Blackboard Systems*. Reading, Mass.: Addison-Wesley.

Journal and magazine articles Use the `@article` class.

Robinson, A. L. 1980. New Ways to Make Microcircuits Smaller. *Science*, 208(4447): 1019–1022.

Hasling, D. W.; Clancey, W. J.; and Rennels, G. 1984. Strategic explanations for a diagnostic consultation system. *International Journal of Man-Machine Studies*, 20(1): 3–19.

Proceedings paper published by a society, press or publisher Use the `@inproceedings` class. You may abbreviate the `booktitle` field, but make sure that the conference edition is clear.

Clancey, W. J. 1984. Classification Problem Solving. In *Proceedings of the Fourth National Conference on Artificial Intelligence*, 45–54. Menlo Park, Calif.: AAAI Press.

Clancey, W. J. 1983. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence (IJCAI-83)*, 556–560. Menlo Park, Calif: IJCAI Organization.

University technical report Use the @techreport class.

Rice, J. 1986. Poligon: A System for Parallel Problem Solving. Technical Report KSL-86-19, Dept. of Computer Science, Stanford Univ.

Dissertation or thesis Use the @phdthesis class.

Clancey, W. J. 1979. *Transfer of Rule-Based Expertise through a Tutorial Dialogue*. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Forthcoming publication Use the @misc class with a note="Forthcoming" annotation.

```
@misc(key,
[...],
note="Forthcoming",
)
```

Clancey, W. J. 2021. The Engineering of Qualitative Models. Forthcoming.

ArXiv paper Fetch the BibTeX entry from the "Export BibTeX Citation" link in the arXiv website. Notice it uses the @misc class instead of the @article one, and that it includes the eprint and archivePrefix keys.

```
@misc(key,
[...],
eprint="xxxx.yyyy",
archivePrefix="arXiv",
)
```

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. arXiv:1706.03762.

Website or online resource Use the @misc class. Add the url in the howpublished field and the date of access in the note field:

```
@misc(key,
[...],
howpublished="\url{http://...}",
note="Accessed: YYYY-mm-dd",
)
```

NASA. 2015. Pluto: The 'Other' Red Planet. <https://www.nasa.gov/nh/pluto-the-other-red-planet>. Accessed: 2018-12-06.

For the most up to date version of the AAAI reference style, please consult the *AI Magazine* Author Guidelines at <https://aaai.org/ojs/index.php/aimagazine/about/submissions#authorGuidelines>

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Thank you for reading these instructions carefully. We look forward to receiving your electronic files!

References

Clancey, W. J. 1979. *Transfer of Rule-Based Expertise through a Tutorial Dialogue*. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Clancey, W. J. 1983. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence (IJCAI-83)*, 556–560. Menlo Park, Calif: IJCAI Organization.

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NASA. 2015. Pluto: The 'Other' Red Planet. <https://www.nasa.gov/nh/pluto-the-other-red-planet>. Accessed: 2018-12-06.

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Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. arXiv:1706.03762.